# Windows Server Basic & Installation

**Weekly Review and Practice Practical (2 hours):Set up a basic AD environment with user accounts, GPOs, and shared resources. Task: Submit a report summarizing the setup with screenshots and configurations. Checkpoint:Quiz: Active Directory concepts, GPOs, and file sharing. Practical Evaluation: Configure AD with GPOs and shared folders.**

**1. Install Windows Server (any version) on a virtual machine using VirtualBox or VMware, and take a screenshot of the successful installation desktop.**

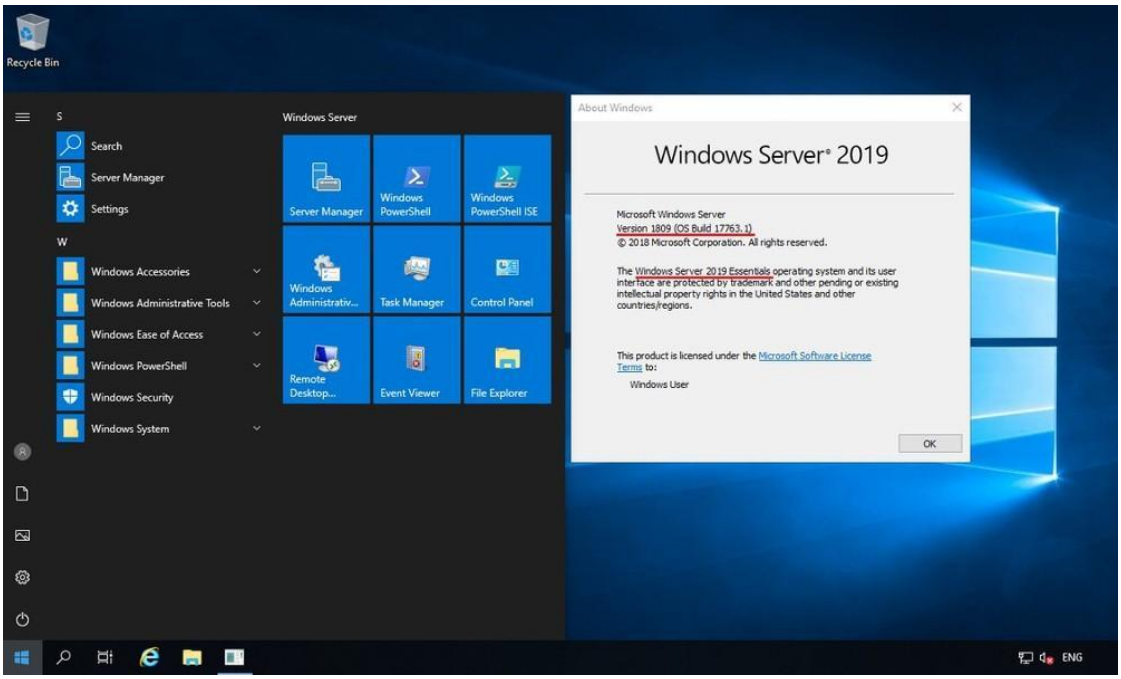
**ANS:**

**Step 1: Create a virtual machine**

1. Open VMware Workstation.
2. Click Create a New Virtual Machine.
3. Select Typical (Recommended).
4. Select the Windows Server ISO file.
5. Choose:
    - Guest OS: Microsoft Windows
    - Version: Windows Server 2019
6. Name the virtual machine MUSIC-SERVER.
7. Allocate resources:

| Hardware | Recommended |
|---|---|
| RAM | 4 GB minimum |
| CPU | 2 cores |
| Hard disk | 60 GB |
| Network | NAT or Host-only |

8. Click Finish.
9. Start the virtual machine.




**Step 2: Install Windows Server**

1. Select your language and keyboard settings.
2. Click Install Now.
3. Select Windows Server 2019 Standard (Desktop Experience).
4. Accept the license agreement.
5. Select Custom Installation.
6. Choose the virtual hard disk.
7. Wait for installation to complete.
8. Set the Administrator password.
9. Log in using the Administrator account.

**Step 3: Rename the server**

1. Open Server Manager.
2. Click Local Server.
3. Click the current computer name.
4. Click Change.
5. Enter: MUSIC-SERVER
6. Restart the server.

**Step 4: Configure a static IP address**

Open:

Control Panel → Network and Sharing Center → Change adapter settings

Right-click Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4).

Enter:

| **Setting** | **Value** |
|---|---|
| IP address | 192.168.128.101 |
| Subnet mask | 255.255.255.0 |
| Default gateway | 192.168.128.1 |
| Preferred DNS | 192.168.128.101 |

Click OK.

**2. Create a new Active Directory domain called "musicstream.local" and add two user accounts: one for 'arijit.singh' and one for 'neha.kakkar'. Capture screenshots of the AD Users and Computers console showing both accounts.**

**ANS:**

**Step 1: Install Active Directory Domain Services**

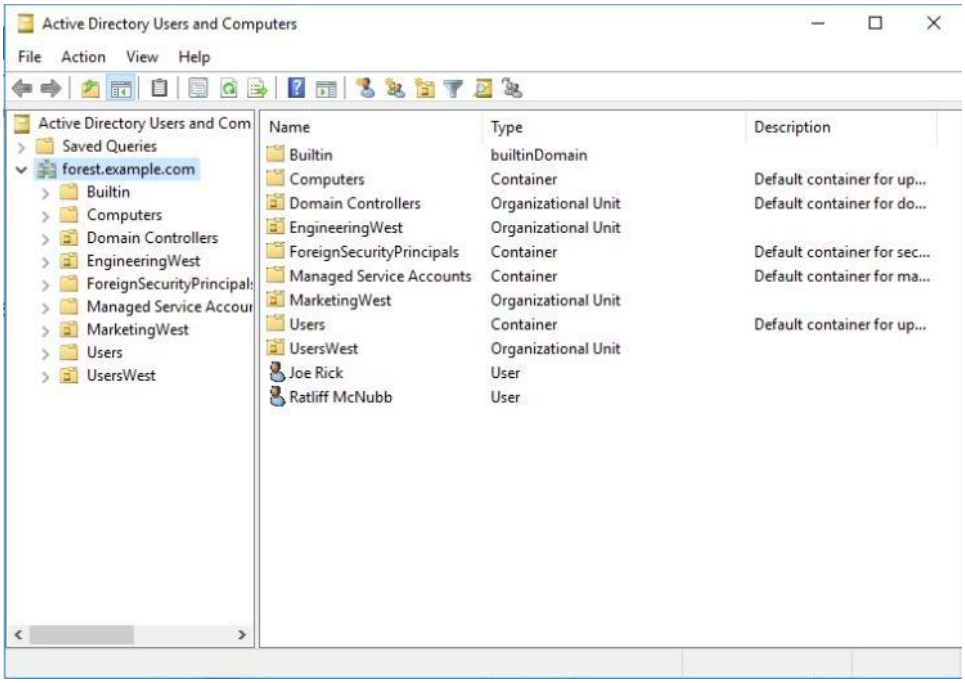
1. Open Server Manager.
2. Click Manage.
3. Select Add Roles and Features.
4. Click Next until Server Roles.
5. Select Active Directory Domain Services.
6. Click Add Features.
7. Click Next.
8. Click Install.
9. Wait until the installation finishes.

**Step 2: Promote the server to a Domain Controller**

1. In Server Manager, click the notification flag.
2. Select Promote this server to a domain controller.
3. Select Add a new forest.
4. Enter the Root domain name: musicstream.local
5. Click Next.
6. Select the following:
    - Domain Name System (DNS) Server
    - Global Catalog
7. Set the Directory Services Restore Mode (DSRM) password.
8. Click Next through the remaining configuration screens.
9. Click Install.
10. Wait for the server to restart.

Your Active Directory domain is now:

musicstream.local




**Step 3: Open Active Directory Users and Computers**

1. Open Server Manager.
2. Click Tools.
3. Select Active Directory Users and Computers.
4. Expand musicstream.local.

**Step 4: Create a Users OU**

Do not use the default Users container. Create a separate OU so you can link your GPO directly to it.

1. Right-click musicstream.local.
2. Select New → Organizational Unit.
3. Enter: Users
4. Click OK.

**Step 5: Create the first user – Arijit Singh**

1. Right-click the Users OU.
2. Select New → User.
3. Enter:

| **Field** | **Value** |
|---|---|
| First name | Arijit |
| Last name | Singh |
| User logon name | arijit.singh |

4. Click Next.
5. Set a password.
6. Uncheck User must change password at next logon (for easier lab testing).
7. Select Password never expires only if permitted by your lab policy.
8. Click Next → Finish.

**Step 6: Create the second user – Neha Kakkar**

Repeat the same process.

| **Field** | **Value** |
|---|---|
| First name | Neha |
| Last name | Kakkar |
| User logon name | neha.kakkar |

**3. Set up a Group Policy Object (GPO) that prevents users from accessing Control Panel, and link it to the 'Users' OU you created. Show before-and-after screenshots of the policy in effect for one of your test users.**

**ANS:**

**Step 1: Open Group Policy Management**

1. Open Server Manager.
2. Click Tools.
3. Select Group Policy Management.
4. Expand:
    - Forest: musicstream.local
    - Domains
    - musicstream.local

**Step 2: Create a new GPO**

1. Right-click the Users OU.
2. Select Create a GPO in this domain, and Link it here.
3. Enter the GPO name:

   Block_Control_Panel

4. Click OK.

**Step 3: Edit the GPO**

1. Right-click Block_Control_Panel.
2. Click Edit.
3. Navigate to:

   User Configuration
   └── Policies
   └── Administrative Templates
   └── Control Panel

4. Find the policy:

Prohibit access to Control Panel and PC settings

5. Double-click it.
6. Select Enabled.
7. Click Apply.
8. Click OK.

**Step 4: Verify the GPO**

On your Windows 10 client:

1. Make sure the client is joined to musicstream.local.
2. Log in using the Arijit Singh domain account.
3. Open Command Prompt as administrator.
4. Run:

gpupdate /force

5. Sign out and sign in again as arijit.singh.
6. Press Windows + R.
7. Enter: control
8. Press Enter.

**Step 5: Test before and after applying the policy**

Before applying GPO:

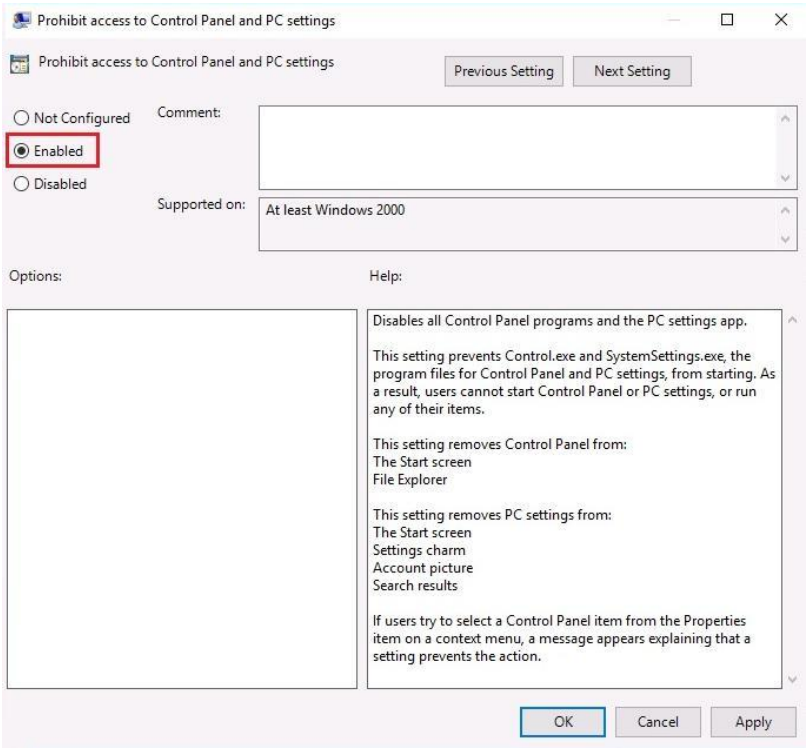
- Log in as arijit.singh.
- Open Run.
- Type control.
- Verify that Control Panel opens.
- Capture a screenshot.

After applying GPO:

- Run gpupdate /force.
- Sign out and sign back in.
- Type control again.
- Verify that access is blocked.
- Capture another screenshot.

**Step 6: Verify the applied policy**

Run: gpresult /r



**4. Share a folder named 'Playlists' on your Windows Server so that only the 'arijit.singh' user can read and write files, but 'neha.kakkar' can only read. Demonstrate this by logging in as each user and attempting to create or edit a file in the shared folder. Hint: Use NTFS permissions and sharing settings to control access for each user.**

**ANS:**

**Step 1: Create the Playlists folder**

1. Open File Explorer on Windows Server.
2. Open the C: drive.
3. Create a new folder named:

   Playlists

   The folder path should be:

   C:\Playlists

**Step 2: Configure Sharing Permissions**

1. Right-click C:\Playlists.
2. Select Properties.
3. Open the Sharing tab.
4. Click Advanced Sharing.
5. Check Share this folder.
6. Enter the Share name: Playlists
7. Click Permissions.

Configure these permissions:

| Group/User | Allow |
|---|---|
| Domain Users | Change, Read |
| Administrators | Full Control |

If Everyone is present, remove it.

Click Apply → OK.

Why Domain Users gets Change permission: This is the share-level permission. The more restrictive NTFS permissions configured next will determine what each user can actually do.

**Step 3: Configure NTFS Security Permissions**

1. Open the folder Properties.
2. Click the Security tab.
3. Click Advanced.
4. Click Disable inheritance.
5. Select Convert inherited permissions into explicit permissions.
6. Remove unnecessary broad entries such as Users or Authenticated Users if they grant access beyond your intended permissions.
7. Keep:
    - SYSTEM – Full Control
    - Administrators – Full Control
8. Click Add.

**Step 4: Give Arijit Singh Read and Write permissions**

1. Click Select a principal.
2. Enter: MUSICSTREAM\arijit.singh
3. Click Check Names.
4. Click OK.
5. Select the following permissions:
    - Modify
    - Read & execute
    - List folder contents
    - Read
    - Write
6. Apply to: This folder, subfolders and files.
7. Click OK.

**Step 5: Give Neha Kakkar Read-only permissions**

1. Click Add.
2. Select a principal.
3. Enter: MUSICSTREAM\neha.kakkar
4. Click Check Names.
5. Click OK.
6. Select only:
    - Read & execute
    - List folder contents
    - Read
7. Apply to: This folder, subfolders and files.
8. Click OK.
9. Click Apply → OK.

**Step 6: Access the shared folder**

On your Windows 10 client:

1. Log in as arijit.singh.
2. Press Windows + R.
3. Enter: \\MUSIC-SERVER\Playlists
4. Press Enter.

If the server name does not resolve, use:

\\192.168.128.101\Playlists

**Step 7: Test Arijit Singh permissions**

While logged in as arijit.singh:

1. Open the Playlists folder.
2. Right-click → New → Text Document.
3. Create: Arijit_Playlist.txt
4. Open the file.
5. Write: My favourite songs are available here.
6. Save the file.
7. Reopen it and edit the contents.
8. Save again.

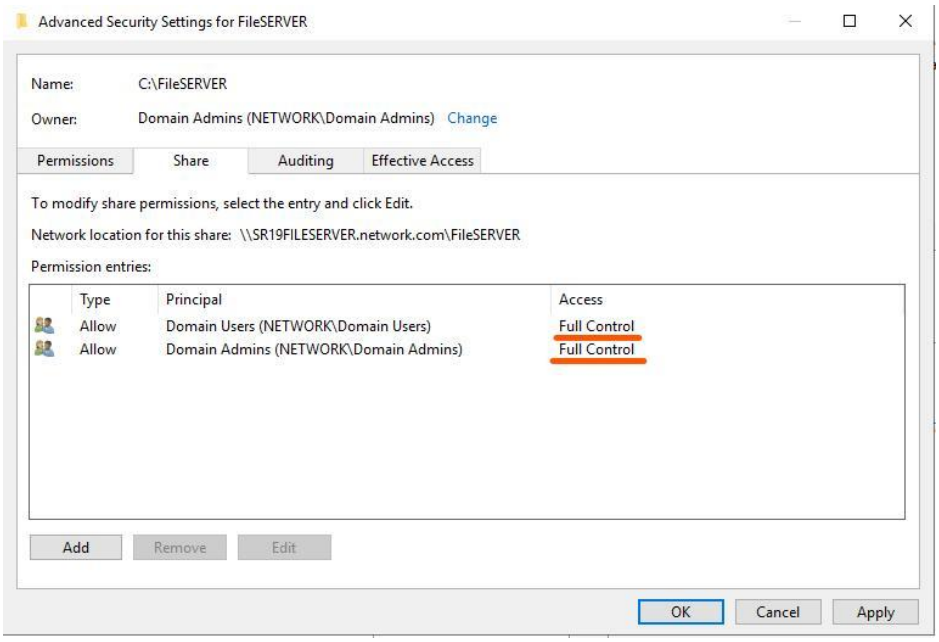
**Step 8: Test Neha Kakkar permissions**

Sign out and log in as neha.kakkar.

Open: \\MUSIC-SERVER\Playlists

Test the following:

1. Open Arijit_Playlist.txt.
2. Read its contents.
3. Try creating a new text document.
4. Try editing and saving Arijit's existing file.



**5. Write a short report (1-2 pages) summarizing your domain, user, GPO, and shared folder setup. Include at least 3 screenshots and describe one challenge you faced and how you solved it.**

**ANS:**

## 1. Introduction

The objective of this practical assignment was to install Windows Server 2019, configure an Active Directory domain, create domain user accounts, implement Group Policy restrictions and configure a shared folder with different access permissions for different users.

The practical was performed using VMware Workstation and Windows Server 2019.

## 2. Active Directory Domain Configuration

First, Windows Server 2019 was installed on a virtual machine. The server was renamed to MUSIC-SERVER and assigned a static IP address of 192.168.128.101.

The Active Directory Domain Services (AD DS) role was installed through Server Manager. After installation, the server was promoted to a Domain Controller by creating a new forest with the domain name **musicstream.local**.

A new Organizational Unit (OU) named Users was created. Two domain user accounts were created inside this OU:

- arijit.singh
- neha.kakkar

Both accounts were successfully created and displayed in the Active Directory Users and Computers console.

## 3. Group Policy Configuration

A new Group Policy Object named **Block_Control_Panel** was created and linked to the Users OU.

The following policy was configured:

User Configuration → Policies → Administrative Templates → Control Panel → Prohibit access to Control Panel and PC settings.

The policy was enabled to prevent users from accessing Control Panel.

Before applying the policy, Control Panel was accessible. After applying the GPO and updating Group Policy using gpupdate /force, access to Control Panel was successfully restricted for the test user.

The applied policy was verified using the gpresult /r command.

**4. Shared Folder Configuration**

A shared folder named **Playlists** was created on the Windows Server at:

C:\Playlists

The folder was shared over the network using the share name Playlists.

Both sharing permissions and NTFS security permissions were configured to provide different access levels to the two domain users.

The following permissions were assigned:

- **arijit.singh:** Read, Write and Modify permissions.
- **neha.kakkar:** Read-only permissions.

The shared folder was accessed from a Windows client using:

\\MUSIC-SERVER\Playlists

Arijit Singh successfully created and edited text files in the shared folder. Neha Kakkar could read existing files but was unable to create or modify files.

**5. Challenges Faced and Solutions**

One challenge during this practical was configuring the correct permissions for the shared folder. Initially, different Windows permission settings could affect the users' ability to create or modify files.

This was resolved by configuring both Share Permissions and NTFS Security Permissions. Arijit Singh was assigned Modify permission, while Neha Kakkar was assigned Read-only permission. After testing both accounts, the required access restrictions worked correctly.

**6. Conclusion**

In this practical assignment, Windows Server 2019 was successfully configured as a Domain Controller for the musicstream.local domain. Two Active Directory user accounts were created, and a Group Policy Object was implemented to restrict Control Panel access.

Additionally, the Playlists shared folder was configured with different permissions for both users. The practical demonstrated how Active Directory, Group Policy and NTFS permissions can be used to manage users and secure shared network resources in a Windows Server environment.

**Result:** All five assignment tasks were successfully completed and verified.